

Liwei Hu

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Research Interests

Computational inverse problems; structure-aware regularization; uncertainty quantification; numerical optimization; geophysical and Earth-system applications.

Education

PhD, Applied Mathematics

UNIVERSITY OF BOLOGNA

- Advisor: Fabiana Zama
- Co-Advisors: Lisa Borgatti, Germana Landi, Davide Donati, Giacomo Titti
- Doctoral Program: EIT4SEMM

Bologna, Italy

2023–present

MS, Applied Mathematics

UNIVERSITY OF BOLOGNA

- Thesis Advisor: Fabiana Zama

Bologna, Italy

2020–2023

BS, Mathematics

UNIVERSITY OF BOLOGNA

Bologna, Italy

2017–2020

Research Experience

Case Western Reserve University

MENTORS: DANIELA CALVETTI, ERKKI SOMERSALO

- Role: Visiting PhD Researcher
- Topic: Bayesian scientific computing

Cleveland, OH

Sep - Dec 2025

University of California, Berkeley

MENTOR: ROLAND BÜRGMANN

- Role: Visiting PhD Researcher
- Topic: Bayesian inverse problems for Earth surface dynamics

Berkeley, CA

Apr - Jul 2025

Publications

† Authors listed in alphabetical order

PUBLISHED

†Borgatti L., Donati D., **Hu L.**, Landi G., Zama F. (2026). Estimating landslide thickness through domain-based regularization. *Journal of Computational and Applied Mathematics*, 476, 117073.

Titti G., **Hu L.**, Festi P., Elia L., Borgatti L., Lombardo L. (2025). An updated version of the SZ-plugin: From space to space–time data-driven modeling in QGIS. *International Journal of Applied Earth Observation and Geoinformation*, 142, 104679.

Spinelli G.V., Evangelista D., **Hu L.**, Zama F. (2025). Neural network-based inversion of NMR dispersion profiles for enhanced analysis of food systems. *Neural Computing & Applications*, 37, 3957–3972.

†Borgatti L., Donati D., **Hu L.**, Landi G., Zama F. (2024). A regularization method for landslide thickness estimation. *Journal of Imaging*, 10(12): 314.

SUBMITTED

Hu L., Calvetti D., Landi G., Somersalo E., Zama F. (2026). Bayesian variational approach to landslide thickness estimation. Submitted to *Inverse Problems*.

MANUSCRIPTS IN PREPARATION

Hu L., Calvetti D., Landi G., Somersalo E., Zama F. (2026). Uncertainty quantification in landslide thickness estimation. Manuscript in preparation for *SIAM/ASA Journal on Uncertainty Quantification*.

Hu L., Bürgmann R., Xu Y., Hu X., Donati D., Landi G., Zama F., Borgatti L. (2026). Inferring landslide slip surface using kinematic elements. Manuscript in preparation for *Journal of Geophysical Research: Earth Surface*.

Presentations _____

* presenting author

CONTRIBUTED PRESENTATIONS

Hu L.* , Calvetti D., Landi G., Somersalo E., Zama F. (2026). *Regularization for landslide thickness estimation via finite element methods*. Oral presentation, SIAM Annual Meeting 2026, Cleveland, OH.

Hu L.* , Bürgmann R., Xu Y., Hu X., Donati D., Landi G., Zama F., Borgatti L. (2025). *Locating landslide slip surfaces using kinematic elements: A case study for Slumgullion landslide*. Geoburst oral presentation, AGU Annual Meeting 2025, New Orleans, LA.

Borgatti L., Donati D., Hu L.* , Landi G., Zama F. (2025). *A regularization method for landslide thickness estimation*. Departmental seminar, Department of Mathematics, University of Bologna, Bologna, Italy.

Hu L.* , Donati D., Landi G., Zama F., Borgatti L. (2024). *Development and application of an enhanced inversion algorithm for the reconstruction of the basal surface of landslides*. PhD Day oral presentation, Italian Geological Society-Italian Society of Mineralogy and Petrology 2024 Congress: Geology for sustainable management of our Planet, Bari, Italy.

Borgatti L., Donati D., Hu L.* , Landi G., Zama F. (2024). *A regularisation approach to landslide thickness estimation*. Oral presentation, Workshop on Advanced Numerical Analysis for Imaging: from theory to industry, Naples, Italy.

POSTER PRESENTATIONS

Hu L.* , Bürgmann R., Xu Y., Hu X., Donati D., Landi G., Zama F., Borgatti L. (2025). *Locating landslide slip surfaces using kinematic elements: A case study for Slumgullion landslide*. Poster presentation, AGU Annual Meeting 2025, New Orleans, LA.

Borgatti L., Donati D., Hu L.* , Landi G., Zama F. (2024). *An inverse problem from engineering geology: a regularisation approach for landslide thickness estimation*. Poster presentation, 4th IMA Conference on Inverse Problems from Theory to Application, Bath, UK.

Teaching Experience _____

Fall 2025	Directed reading program in discrete inverse problems , Graduate Mentor, Department of Mathematics, Applied Mathematics, and Statistics, Case Western Reserve University	Cleveland, OH
Spring 2025	Numerical Differential Equations , Teaching Assistant, Department of Mathematics, University of Bologna	Bologna, Italy
Fall 2024	Numerical Methods , Teaching Assistant, Department of Mathematics, University of Bologna	Bologna, Italy
Spring 2024	Numerical Differential Equations , Teaching Assistant, Department of Mathematics, University of Bologna	Bologna, Italy
Fall 2023	Numerical Methods , Teaching Assistant, Department of Mathematics, University of Bologna	Bologna, Italy

Awards, Fellowships, & Grants

2025	Marco Polo Mobility Grant , University of Bologna	€ 8100
2023	Emilia-Romagna Region Award , Emilia-Romagna Region	€ 15,000
		€ 56,415 (3
2023–2026	PhD Fellowship , University of Bologna	years; fully funded)